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Air Quality Measurements from the Fresno Supersite

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ABSTRACT

The Fresno Supersite intends to 1) evaluate non-routine monitoring methods, establishing their comparability with existing methods and their applicability to air quality planning, exposure assessment, and health effects studies; 2) provide a better understanding of aerosol characteristics, behavior, and sources to assist regulatory agencies in developing standards and strategies that protect public health; and 3) support studies that evaluate relationships between aerosol properties, co-factors, and observed health end-points. Supersite observables include in-situ, continuous, short-duration measurements of 1) $PM_{2.5}$, PM_{10}, and coarse (PM_{10} minus $PM_{2.5}$) mass; 2) $PM_{2.5}$ SO_4^{-2}, NO_3^-, carbon, light absorption, and light extinction; 3) numbers of particles in discrete size bins ranging from 0.01 to ~10 μm; 4) criteria pollutant gases (O_3, CO, NO_x); 5) reactive gases (NO_2, NO_y, HNO_3, peroxyacetyl nitrate [PAN], NH_3); and 6) single particle characterization by time-of-flight mass spectrometry. Field sampling and laboratory analysis are applied for gaseous and particulate organic compounds (light hydrocarbons, heavy hydrocarbons, carbonyls, polycyclic aromatic hydrocarbons [PAH], and other semi-volatiles), and $PM_{2.5}$ mass, elements, ions, and carbon. Observables common to other Supersites are 1) daily $PM_{2.5}$ 24-hr average mass with Federal Reference Method (FRM) samplers; 2) continuous hourly and 5-min average $PM_{2.5}$ and PM_{10} mass with beta attenuation monitors (BAM) and tapered element oscillating microbalances (TEOM); 3) $PM_{2.5}$ chemical speciation with a U.S. Environmental Protection Agency (EPA) speciation monitor and protocol; 4) coarse particle mass by dichotomous sampler and difference between PM_{10} and $PM_{2.5}$ BAM and TEOM measurements; 5) coarse particle chemical composition; and 6) high sensitivity and time resolution scalar and vector wind speed, wind direction, temperature, relative humidity, barometric pressure, and solar radiation. The Fresno Supersite is coordinated with health and toxicological studies that will use these data in establishing relationships with asthma, other respiratory disease, and cardiovascular changes in human and animal subjects.

IMPLICATIONS

EPA's Supersite program will supply more detailed measurements of particle characteristics than has heretofore been available. The long-term databases developed at these sites will be available to evaluate different measurement technologies, to understand atmospheric processes, and to develop more precise relationships between particle indicators and adverse health effects.

INTRODUCTION

The U.S. Environmental Protection Agency's (EPA) Supersites program[1] intends to operate research-grade air monitoring stations to improve understanding of measurement technologies, source contributions and control strategies, and effects of suspended particles on health. Supersites are being established in seven urban areas within the continental United States: 1) Fresno, CA; 2) Los Angeles, CA; 3) Houston, TX; 4) St. Louis, MO; 5) Pittsburgh, PA; 6) Baltimore, MD; and 7) New York, NY. Supersites are designed to 1) test specific scientific hypotheses appropriate for the monitored airshed and suite of measurements; 2) provide measurements that can be compared and contrasted among the seven urban areas; 3) add value to larger monitoring networks and research studies; and 4) leverage EPA investments with contributions from other agencies. The information derived from these Supersites is expected to complement information from $PM_{2.5}$ and PM_{10} measurement networks operated at community representative (CORE), transport, and background

locations as part of the national $PM_{2.5}$ compliance and IMPROVE (Interagency Monitoring of Protected Visual Environments) networks.

The Fresno Supersite is acquiring advanced air quality measurements related to suspended PM to accomplish the following objectives:

- test and evaluate non-routine monitoring methods, with the intent to establish their comparability with existing methods and determine their applicability to air quality planning, exposure assessment, and health impact determination;
- increase the knowledge base of aerosol characteristics, behavior, and sources so regulatory agencies can develop standards and strategies that protect public health; and
- evaluate relationships between aerosol properties, co-factors, and observed health end-points.

Specific hypotheses have been formulated related to each of these objectives. Measurement methods have been selected to provide the variables needed to test these hypotheses. Supersite measurements are coordinated with health and toxicological studies in the region, specifically a multi-million dollar effort by the state of California to better understand the relationships between air quality and asthma.

Previous studies[2-32] show that emissions, aerosol composition, and meteorology change substantially over the course of a day, between seasons, and between different years in central California, where Fresno is located. It is conjectured that relationships between different measurement methods, aerosol characteristics, and health endpoints depend on these changes. Evaluating these relationships requires frequent sampling over short durations for a multi-year monitoring period.

Although all of the measurement methods applied at Fresno have been used in short-term programs, the Supersite represents the first application of these methods together for multi-year monitoring. The objectives of this paper are to 1) describe the area represented by Supersite monitoring and the historical air quality context for current measurements; 2) specify the variables measured at the Supersite and at surrounding locations with respect to method, sampling period, sampling frequency, and sampling duration; and 3) state the hypotheses to be tested with measurements and data analysis methods needed for that testing.

FRESNO STUDY AREA

Fresno and its sister city of Clovis are located in the center of California's San Joaquin Valley, which encompasses nearly 64,000 km^2 and contains a population in excess of 3 million people. The Fresno metropolitan area is the largest population center for 150 km to the north and south. The population of more than 500,000 is ethnically diverse, with ~51% White, ~36% Hispanic, ~5% African-American, and ~8% Asian. The Fresno area experiences frequent hospitalizations for asthma, ranking 2nd in California for African-American children, 3rd for Hispanic children, and 8th for Caucasian children.

The San Joaquin Valley in which Fresno is situated is bordered on the west by the coastal mountain range and on the east by the Sierra Nevada range.[6] These ranges converge at the Tehachapi Mountains in the southern end of the Valley, nearly 200 km south of Fresno. Weather changes with season throughout the year. Spring often experiences small frontal passages with low moisture content resulting in high winds. Summer meteorology is driven by heating over the Mojave desert that creates a thermal low pressure system and a large pressure gradient between the coast and the desert. Fall is influenced by the Great Basin High, with prolonged periods of slow air movement and limited vertical mixing. Mixing depths and ventilation are low in the morning during all seasons and remain low throughout the day during the winter. Relative humidities are highest in the winter, with low relative humidities in the summer and fall. For spring, summer, and fall, the typical winds are northwesterly, directed along the Valley axis. This is the predominant non-winter wind-flow pattern both during the day and the night, although it is more sluggish during fall.

Central California emission source categories include 1) small to medium-sized point sources (e.g., power stations, incinerators, cement plants, and steam generators); 2) area sources (e.g., fires, wind-blown dust, petroleum extraction operations, cooking, and residential fuel combustion); 3) mobile sources (e.g., cars, trucks, off-road heavy equipment, trains, and aircraft); 4) agricultural and ranching activities (e.g., fertilizers, herbicides, tilling, and livestock); and 5) biogenic sources (e.g., oxides of nitrogen from biological activity in soils and hydrocarbon emissions from plants). Agriculture is the main industry surrounding Fresno, with cotton, alfalfa, corn, safflower, grapes, and tomatoes being the major crops. Cattle feedlots, dairies, chickens, and turkeys constitute most of the animal husbandry in the region. Oil and gas production, refining to the south, waste incineration to the northwest, electrical co-generation at various locations, transportation, commerce, local government, and light manufacturing constitute the remainder of the economy.

$PM_{2.5}$ levels measured from 1991 to 1996 with dichotomous samplers operating every 6th day show annual averages ranging from 18 to 24 $\mu g/m^3$. The highest 24-hr averages ranged from 56 to 93 $\mu g/m^3$ during this period. These exceed the annual (15 $\mu g/m^3$) and 24-hr (65 $\mu g/m^3$) National Ambient Air Quality Standards (NAAQS) for $PM_{2.5}$. The highest $PM_{2.5}$ concentrations are

typically found during winter and fall, with the lowest concentrations occurring during spring and summer.[33] $PM_{2.5}$ constitutes ~80% of PM_{10} during winter and ~50% of PM_{10} during the rest of the year.[6,8,9]

Fresno experiences high NH_4NO_3 levels during winter and sometimes fall, large geological contributions during fall and spring, and medium to high carbon concentrations throughout the year. Organic carbon concentrations are enhanced by vegetative burning and cooking during winter[34] when dispersion is poor. $(NH_4)_2SO_4$ levels average less than 2 μg/m^3 throughout the year. Strader et al.[29] hypothesize that up to 20% of wintertime organic material during some parts of the day may be of secondary origin, owing to multi-day accumulation of organic precursors, but organic to elemental carbon proportions are typically similar to those of primary emissions. This rich variability in aerosol concentration and composition found throughout the year, over multi-day episodes, and even within a day provides the contrasts and extremes needed to stress measurement methods and to evaluate changes in health end-points. The Fresno atmosphere presents substantial measurement challenges caused by multiple area sources, volatile aerosol, and fogs and rain during winter.

SITE DESCRIPTION

The California Air Resources Board (ARB) has operated the site at 3425 First Street (coordinates –119.7727725 °W, 36.78184232 °N), ~5.5 km north-northeast of the downtown commercial district, since 1990. Commercial establishments, office buildings, churches, and schools are located north and south of the Fresno Supersite on First Street, a four-lane artery with moderate traffic levels. Medium-density single-family homes and some apartments are located in the blocks to the east and west of First Street. Sampling inlets are located on a second-story rooftop ~10 m above ground level and ~30 m from the west-side curb of First Street. Other PM sites in the Fresno metropolitan area include the Clovis residential site located ~7 km northeast of the Supersite, the Drummond Street industrial site ~9 km south-southeast, and the southeast Fresno suburban site ~7.5 km southeast.

Two satellite sites[35] are located in the vicinity of the Supersite to determine its zone of representation and the effects of local sources on chemical concentrations. A vehicle-dominated site at Shields Avenue and SR 41 is in a residential area near a four-lane arterial and a freeway on-ramp ~1 km west-southwest of the Supersite and will quantify incremental carbon contributions from directly emitted vehicle exhaust. A residential site near a city park is located ~0.5 km east of the Supersite on a lightly traveled neighborhood street. For most of the year, this site will represent a neighborhood similar to that around the Supersite. In winter, however, Schauer and Cass[36] attributed the large increment in organic carbon near this site to neighborhood wood combustion. A third satellite site located at the Selma Airport, ~24 km south-southeast of the Supersite, is outside of the populated area and provides the ability to separate urban-scale from regional-scale PM contributions.

The Angiola Tower site operated as part of the California Regional $PM_{10}/PM_{2.5}$ Air Quality Study (CRPAQS)[33] is equipped with much of the same measurements as the Fresno Supersite. The Angiola site is located in a flat field ~85 km south-southeast of Fresno with minimal influence from non-urban sources. A 100-m tower is instrumented with micrometeorological and particle monitors to evaluate the vertical mixing and transport of pollutants between major urban areas such as Fresno and Bakersfield. Measurements at the Angiola Tower site will be compared with those from Fresno to evaluate hypotheses about urban- and regional-scale contributions to excessive $PM_{2.5}$ and PM_{10} concentrations.

SUPERSITE MEASUREMENTS

Table 1 specifies the measurements at the Fresno Supersite, the methods applied, sample durations and frequencies, and measurement periods. Simultaneous measurements available from the Angiola Tower site are also indicated, as are measurements at the three satellite sites.

Emphasis is on in-situ, continuous, short-duration measurements of 1) $PM_{2.5}$, PM_{10}, and coarse (PM_{10} minus $PM_{2.5}$) mass; 2) $PM_{2.5}$ SO_4^{-2}, NO_3^-, carbon, light absorption, and light extinction; 3) numbers of particles in discrete size ranges from 0.01 to ~10 μm; 4) criteria pollutant gases (O_3, CO, NO_x); 5) reactive gases (NO_2, NO_y, HNO_3, peroxyacetyl nitrate [PAN], NH_3); and 6) single particle characterization. Field sampling and laboratory analysis are applied for 1) gaseous and particulate organic compounds (light hydrocarbons, heavy hydrocarbons, carbonyls, polycyclic aromatic hydrocarbons [PAH], and other semi-volatiles); 2) $PM_{2.5}$ mass, elements, ions, and carbon; and 3) $PM_{2.5}$ and coarse-particle bioaerosols such as endotoxins, pollens, spores, molds, and fungi.

Observables common to other Supersites are 1) daily $PM_{2.5}$ 24-hr average mass with Federal Reference Method (FRM) samplers; 2) continuous hourly and 5-min average $PM_{2.5}$ and PM_{10} mass with beta attenuation monitors (BAM) and tapered element oscillating microbalances (TEOM); 3) $PM_{2.5}$ chemical speciation with an EPA speciation monitor and protocol; 4) coarse particle mass by dichotomous sampler and difference between PM_{10} and $PM_{2.5}$ BAM and TEOM measurements; 5) coarse particle chemical composition; and 6) high-sensitivity and time-resolved scalar and vector wind speed, wind direction, temperature, relative humidity, barometric pressure, and

Table 1. Summary of air quality and meteorological measurements at the Fresno Supersite.

Observable and Method	Size Range	Avg Time	Frequency	Period
Gases				
Nitrogen oxides (NO/NO_x) (TEI 42 chemiluminescence with internal converter)[a]	gas	5 min	daily	1990 onward[b]
NO_2/PAN (UC Riverside Luminol)	gas	5 min	daily	12/1/00 to 3/31/03
Ozone (API 400 UV absorption)[a]	gas	5 min	daily	1990 onward[b]
Carbon monoxide (Dasibi 3008 infrared gas filter correlation)	gas	5 min	daily	1990 onward[b]
Non-methane hydrocarbons (TEI 55C flame ionization)	gas	1 hr	daily	1990 onward[b]
Reactive nitrogen (NO_y) (TEI 42C chemiluminescence with external converter)[a]	gas	5 min	daily	12/15/99 to 3/31/03
HNO_3 (TEI 42C chemiluminescence with external converters and denuders)[c]	gas	5 min	daily	12/1/00 to 3/31/03
NH_3 (TEI 17C with external stainless steel converter)[c]	gas	5 min	daily	12/1/00 to 3/31/03
Filter Mass and Chemistry				
TSP mass (Andersen hivol w/ quartz filters) and lead	TSP	24 hr	12th day	1990 onward[b]
PM_{10} mass, SO_4^{-2}, NO_3^-, Cl^-, and NH_4^+ (Andersen hivol SSI w/ quartz filters)	<10 µm	24 hr	6th day	1990 onward[b]
$PM_{2.5}$ and coarse mass, elements, endotoxins,[d] spores,[d] molds,[d] and fungi[d] (dichotomous samplers with Teflon filters)	<2.5 µm <10 µm	24 hr	6th day	1990 onward[b]
PM_{10}, endotoxins,[d] spores,[d] molds,[d] and fungi[d] (R + P sequential filter and Burkhard samplers)	<10 µm	24 hr	daily	11/00 to 10/04
$PM_{2.5}$ mass, light absorption, elements, ions, and carbon (two Andersen single-channel FRMs w/ Teflon and quartz filters)	<2.5 µm	24 hr	6th day	7/5/99 to 12/26/00 (method evaluation)
$PM_{2.5}$ mass, elements, ions, and carbon (six-channel Andersen RAAS speciation sampler w/denuders and backup filters)	<2.5 µm	24 hr	6th day	7/5/99 to 12/26/00 (methoc evaluation)
Particle morphology (Airmetrics Minivol w/ polycarbonate filter for scanning electron microscopy)	< ~30 µm	24 hr	6th day	7/5/99 to 12/26/00 (method evaluation)
$PM_{2.5}$ mass, elements, ions, and carbon (two-channel Met One speciation sampler)	<2.5 µm	24 hr	3rd day	2001 onward[b]
$PM_{2.5}$ mass, elements, ions, carbon, and NH_3 (two-channel DRI sequential filter sampler w/ HNO_3 denuders and backup filters; mass on all, chemistry on 100 samples)[a]	<2.5 µm	24 hr	daily	12/1/99 to 1/31/01[e]
$PM_{2.5}$ mass, elements, ions, and carbon (two-channel sequential filter sampler w/ denuders and backup filters)[a]	<2.5 µm	3, 5, and 8 hr (5 samples per day)	daily on episode days	15 episode days 12/1/00 to 1/31/01[e]
Toxic species (metals, chromium VI, aldehydes) (Xontec 920)	<~30 µm	24 hr	12th day	1996 onward[b]
Continuous Particle Mass and Chemistry				
$PM_{2.5}$ mass (50 °C R&P 1400a TEOM)	<2.5 µm	1 hr	daily	7/5/99 to 3/31/03
PM_{10} mass (50 °C R&P 1400a TEOM)	<10 µm	1 hr	daily	7/5/99 to 3/31/03
$PM_{2.5}$ mass (ambient temperature Met One 1020 BAM)[a]	<2.5 µm	1 hr	daily	5/15/99 onward[b]
PM_{10} mass (ambient temperature Met One 1020 BAM)[a]	<10 µm	1 hr	daily	5/15/99 onward[b]
$PM_{2.5}$ NO_3^- (R&P/ADI flash volatilization with NO_x detector)[c]	<2.5 µm	10 min	daily	9/23/99 to 3/31/03
$PM_{2.5}$ SO_4^{-2} (ADI flash volatilization with SO_2 detector)[48]	<2.5 µm	10 min	daily	9/23/99 to 10/28/99 2/7/00 to 3/31/03
$PM_{2.5}$ organic and elemental carbon (R&P Series 5400 thermal evolution, OC at 275 °C, EC at 600 °C)[a]	<2.5 µm	30 min	daily	12/15/99 to 3/31/03
$PM_{2.5}$ organic and elemental carbon (GreenTek thermal/optical monitor)	<2.5 µm	30 min	daily	4/1/01 to 3/31/03
Individual particle size and chemistry (UC Riverside time-of-flight spectrometer)	<10 µm	5 min	daily on episode days	12/1/00 to 1/31/01[e]
Particle-bound PAH (EcoChem Analytics PAS2000 w/UV radiation and photoelectric aerosol sensors)	<1 µm	5 min	daily	9/30/99 to 3/31/03

Table 1. Continued.

Observable and Method	Size Range	Avg Time	Frequency	Period
Organic Gases and Particles				
Toxic hydrocarbons (C_2–C_{12}) (Xontec 910 canister sampler)	gas	24 hr	12th day	1995 onward
Carbonyls (Xontec 925 2,4-dinitrophenylhydrazine [DNPH] cartridge sampler)[a]	gas	24 hr summer (4 samples per day)	12th day 3rd day	1995 onward[b] 1995 onward[b]
Carbonyls (DRI sequential sampler with DNPH cartridge)[a]	gas	5–8 hr (4 samples per day)	daily for episodes	15 episode days 12/1/00 to 1/31/01[e]
Light hydrocarbons (C_2–C_{12}) (canister sampler)[a]	gas	5–8 hr (4 samples per day)	daily for episodes	15 episode days 12/1/00 to 1/31/01[e]
Heavy hydrocarbons (C_{10}–C_{20}) (TENAX sampler)[a]	gas	5–8 hr (4 samples per day)	daily for episodes	15 episode days 12/1/00 to 1/31/01[e]
$PM_{2.5}$ organic compounds (DRI sequential sampler with Teflon-coated glass-fiber/PUF/XAD filters)[a]	<2.5 µm	5–8 hr (4 samples per day)	daily for episodes	15 episode days 12/1/00 to 1/31/01[e]
	<2.5 µm	24 hr	6th day	7/5/00 to 8/28/00
$PM_{2.5}$ organic compounds (Airmetrics Minivol w/ Teflon-coated glass-fiber filters) (aggregate 60 samples for organic compound analysis)[a]	<2.5 µm	24 hr	6th day	2/1/00 to 1/31/01[e]
Continuous Light Scattering				
Ambient particle light scattering (Optec NGN2 ambient-temperature nephelometer at 550 nm)	<~30 µm	5 min	daily	5/15/99 to 3/31/03
Total particle light scattering (Radiance M903 nephelometer with smart heater at 530 nm)[a]	<~30 µm	5 min	daily	2/15/00 to 3/31/03
$PM_{2.5}$ particle light scattering (Radiance M903 nephelometer with smart heater at 530 nm)	<2.5 µm	5 min	daily	4/1/00 to 3/31/03
Total particle light scattering (GreenTek GT640A photometer at 780 nm)	<2.5 µm	5 min	daily	1/1/00 to 3/31/03
Total particle light scattering (TSI DustTrak 8520 photometer at 780 nm)	<2.5 µm	5 min	daily	5/15/99 to 3/31/03
Continuous Light Absorption				
Coefficient of haze (RAC 205019-1 paper tape sampler)	<~30 µm	2 hr	daily	1990 onward[b]
Single-wavelength light absorption (McGee AE14U aethalometer at 880 nm)[a]	<2.5 µm	5 min	daily	5/15/99 to 3/31/03
Seven-wavelength light absorption (Andersen AE30S multi-color [350, 450, 570, 590, 615, 660, 880, and 950 nm] aethalometer)	<2.5 µm	5 min	daily	5/15/99 to 3/31/03
Particle Sizes				
Ultrafine particle number by size in 50 size fractions (TSI 3936L10 S scanning mobility particle sizer)[a]	0.010–0.5 µm, 50 bins	5 min	daily	12/15/99 to 3/31/03
Fine particle number in 8 size fractions (PMS Lasair 1003 optical particle counter)[a]	0.1–2 µm (<0.1, <0.2, <0.3, <0.4, <0.5, <0.7, <1.0, and <2.0 µm)	5 min	daily	11/1/99 to 3/31/03
Aerodynamic particle sizer (TSI 3926)	0.3–10 µm	5 min	daily	10/1/00 to 3/31/03
Coarse particle size distribution in 16 size fractions (Climet CI-500 optical particle counter)[a]	0.3–10 µm (<0.3, <0.4, <0.5, <0.63, <0.8, <1.0, <1.3, <1.6, <2.0, <2.5, <3.2, <4.0, <5.0, <6.3, <8.0, and <10 µm)	5 min	daily	12/15/99 to 3/31/03
Mass and ion size distribution in 8 size fractions (MOUDI with Teflon filters for mass and ions)	0.054–10 µm (<0.054, <0.105, <0.148, <0.37, <0.54, <1.0, <1.8, <2.5, and <10 µm)	5–8 hr	daily for episodes	15 episode days 12/1/00-1/31/01[e]

Table 1. Continued.

Observable and Method	Size Range	Avg Time	Frequency	Period
Carbon size distribution in 8 size fractions (MOUDI with aluminum filters for organic and elemental carbon)	0.054–15 μm (<0.054, <0.105, <0.148, <0.37, <0.54, <1.0, <1.8, <2.5, and <10 μm)	5–8 hr	daily for episodes	15 episode days 12/1/00-1/31/01[e]
Meteorology				
Wind speed/direction (Met One 05305L high-sensitivity wind vane and anemometer)[a]	NA[f]	5 min	daily	5/15/99 onward[e]
Temperature (Met One CS500L platinum resistance sensor)[a]	NA[f]	5 min	daily	5/15/99 onward[e]
Relative humidity (Met One CS500L capacitance sensor)[a]	NA[f]	5 min	daily	5/15/99 onward[e]
Solar radiation (Met One LI200X-L pyranometer)[a]	NA[f]	5 min	daily	9/15/99 onward[e]
Atmospheric pressure (Met One piezofilm sensor)[a]	NA[f]	1 hr	daily	9/15/99 onward[e]
Data Acquisition and Processing				
Campbell Scientific 24-input analogue data logger with modem dialup	NA[f]	all times	daily	5/15/99 onward[e]
PC-LABVIEW serial data logger with modem dialup[a]	NA[f]	all times	daily	12/1/99 onward[e]

Notes: [a]These ground-level measurements will also be taken at the non-urban Angiola site established by the CRPAQS from February 1, 2000, through January 31, 2001, and during pollution episodes. This site is located 100 km south of Fresno in a flat area of the San Joaquin Valley surrounded by agricultural fields dominated by cotton and alfalfa. Simultaneous measurements from Angiola will be used with those from the Fresno Supersite to evaluate hypotheses about measurement equivalence in the absence of fresh urban emissions and to separate urban from non-urban contributions to the concentrations of measured observables. CRPAQS episodic measurements at Angiola will be taken at the same time as those acquired at Fresno.

[b]Part of the California ARB's compliance monitoring network.

[c]Measurements at Angiola are available from December 1, 2000, through January 31, 2001.

[d]Part of the Particulate Air Pollution and the Natural History of Childhood Asthma study sponsored by ARB.

[e]Measurements from CRPAQS. Three to five wintertime episodes of 4–8 days duration will be monitored for a total of 15 days between December 1, 2000, and January 31, 2001, based on a forecast of high PM2.5 concentrations under clear sky stagnation and stagnation with fog conditions.

[f]Not applicable.

solar radiation. Satellite sites are equipped with nephelometers operating continuously and with Minivol Teflon and Quartz filter samplers operating for 24 hr every 6th day to quantify mass, elemental, ion, and carbon concentrations. The Radiance Research nephelometers are equipped with "smart heaters" that operate only when humidity exceeds 65%. This minimizes loss of volatile NO_3 while removing interference from aerosol liquid water.

Quality assurance (QA)[1,37-39] plans describe measurements, quality control (QC), and QA activities. They specify Data Quality Objectives[40] to the extent that they apply to a research study and are supported by Standard Operating Procedures (SOPs) describing each measurement process.[41,42]

QA activities for the Fresno Supersite include 1) assembling, reviewing, and archiving SOPs; 2) summarizing QC and QA procedures with measurement descriptions in QA plans; 3) specifying primary, calibration, performance test, and audit standards; 4) specifying data reporting conventions; 5) conducting system audits of field, laboratory, and data management systems; 6) conducting performance audits of field and laboratories; 7) preparing data qualification statements; and 8) reporting QA results.

Several types of standards are needed for calibration, auditing, and performance tests. Development and testing of these standards are important components of Supersite research. Table 2 identifies primary and transfer standards and the frequency of application for calibration, performance testing, and auditing. Methods for presenting these standards depend on the instrument. Flow rates are relatively simple to verify, but evaluating continuous monitor response to particle size is impractical under field conditions. Gas and meteorological monitors used in compliance networks have common procedures and standards for calibration and auditing. Calibration, performance testing, and auditing methods for laboratory operations are largely based on the preparation of standard solutions from mineral salts. The National Institute of Standards and Technology (NIST) does not yet provide these types of standards. NIST and EPA standard solutions in a large range of concentrations are available commercially for inorganic monoatomic and polyatomic ions. Performance testing for particle size and chemical speciation methods are still under development, and one of the research goals is to devise practical standards and methods to evaluate accuracy, precision, and validity of Supersite measurements. Performance test measurements

Table 2. Primary, calibration, performance test, and performance audit standards and frequencies for Fresno measurements.

Observable	Expected Precision	Primary Standard	Calibration Standard	Calibration Frequency	Performance Test Standard	Performance Test Frequency	Performance Audit Standard	Performance Audit Frequency
TSP Minivol sampler flow rates	±15%	NIST-certified bubblemeter	Mass flowmeter/ bubblemeter	Quarterly	Calibrated rotameter	Monthly	Mass flowmeter	5 times over 3 years
TSP and PM_{10} hivol sampler flow rates	±5%	Spirometer	Calibrated orifice/rootsmeter	Quarterly	Calibrated orifice	Monthly	Calibrated orifice/rootsmeter	Yearly
$PM_{2.5}$ and PM_{10} FRM, dichotomous, speciation, and toxic sampler, and MOUDI flow rates	±5%	NIST-certified bubblemeter	Mass flowmeter/ bubblemeter	Quarterly	Calibrated bubblemeter	Monthly	Mass flowmeter	Yearly
$PM_{2.5}$ canister, DNPH, TENAX, organic, and inorganic aerosol sequential speciation sampler flow rates	±5%	NIST-certified Vol-U-Met	Calibrated rotameter	Beginning and end of sampling period	Calibrated rotameter	Monthly	Dry test meter	During sampling period
TEOM and BAM flow rates	±10%	NIST-certified bubblemeter	Mass flowmeter/ bubblemeter	Quarterly	Internal flow check	Weekly	Mass flowmeter	5 times over 3 years
O_3 by ultraviolet absorption	±10%	ARB primary UV photometer	Dasibi 1003PC UV photometer	Quarterly or when out of spec	Span with internal ozone generator and zero with scrubbed air	Daily	Dasibi 1008PC with temperature and pressure adjustments	Yearly
CO by infrared gas filter correlation	±10%	NIST-traceable CO	Certified CO w/dynamic dilution	Quarterly or when out of spec	Span with certified CO and zero with scrubbed air	Daily	Certified CO w/dynamic dilution	Yearly
NMHC flame ionization detection	±10%	NIST-traceable HC mixture	Certified HC gas dilution	Quarterly or when out of spec	Span with certified HC and zero with scrubbed air	Daily	Certified HC gas dilution	Yearly
NO/NO_x by chemiluminescence	±10%	NIST-traceable NO	Certified NO w/dynamic dilution and gas-phase titration	Quarterly or when out of spec	Span with certified NO and zero with scrubbed air	Daily	Certified NO w/dynamic dilution and gas-phase titration	Yearly
NO_y, HNO_3, NO_2, PAN by chemiluminescence and luminol	±20%	NIST-traceable NO	Certified NO w/dynamic dilution and gas-phase titration	Quarterly or when out of spec	Span with certified NO and HNO_3 perm tube and zero with scrubbed air	Daily for NO Weekly for HNO_3	Certified NO w/dynamic dilution and gas-phase titration	3 times over 3 years
NH_3 by chemiluminescence	±20%	NIST-traceable NH_3 mixture	Certified NH_3 w/dynamic dilution	Quarterly or when out of spec	Span with certified NH_3 and zero with scrubbed air	Daily	Certified NH_3 w/dynamic dilution	3 times over 3 years
PM_{10} and $PM_{2.5}$ mass by TEOM	±10%	Certified laboratory weights	Factory calibration	When out of spec	Weighed filter stubs	Quarterly	Weighed filter stubs	5 times over 3 years
PM_{10} and $PM_{2.5}$ mass by BAM	±10%	Certified laboratory weights	Factory calibration	When out of spec	Weighed thin films	Quarterly	Weighed thin films	5 times over 3 years
$PM_{2.5}$ NO_3^- by flash volatilization	±15%	NIST-traceable NO	Certified NO w/dynamic dilution and NO_3^- solution	Quarterly	Span with certified NO and zero with scrubbed air	Weekly	Certified NO w/dynamic dilution and NO_3 solution	5 times over 3 years
$PM_{2.5}$ SO_4^{-2} by flash volatilization	±15%	NIST-traceable SO_2	Certified SO_2 w/dynamic dilution SO_4^{-2} solution	Quarterly	Span with certified SO_2 and zero with scrubbed air	Monthly	Certified SO_2 w/dynamic dilution and SO_4^{-2} solution	5 times over 3 years
$PM_{2.5}$ organic and elemental carbon by thermal evolution	±15%	NIST-certified CO_2	Certified CO_2 w/dynamic dilution	Quarterly	Span with certified CO_2 and zero with nitrogen	Weekly	Certified CO_2 w/dynamic dilution	5 times over 3 years
Particle-bound polycyclic aromatic hydrocarbons by photoionization	±20%	Under development	Factory calibration	When out of spec	Internal electronic checks	Weekly	Collocated comparison with filter/PUF/XAD samples	Under development
Light scattering by nephelometers	±10%	HFC-134a refrigerant	HFC-134a, pure CO_2, and particle-free air	When out of spec	Pure CO_2 and particle-free air	Weekly	HFC-134a, pure CO_2, and particle-free air	5 times over 3 years

Table 2. Continued.

Observable	Expected Precision	Primary Standard	Calibration Standard	Calibration Frequency	Performance Test Standard	Performance Test Frequency	Performance Audit Standard	Performance Audit Frequency
Individual particle size and chemistry by time-of-flight mass spectrometer	±20%	Not available	Under development	Under development	Under development	Under development	Under development	Under development
Light absorption and coefficient of haze by light transmission	±5–20%	Neutral density filters	Factory calibration	When out of spec	Internal electronic checks and neutral density filters	Weekly and monthly	Neutral density filters	Yearly
Ultrafine particle number by scanning mobility particle sizer	±15%	Collocated differential mobility analyzer	Factory calibration	Yearly	Internal electronic checks, collocated comparison	Daily and monthly	Certified particle sizes and voltage tests	5 times over 3 years
Fine and coarse particle number by optical particle counter and aerosol particle sizer	±15%	Vibrating orifice aerosol generator	Polystyrene latex suspension	Yearly	Internal electronic checks, collocated comparison	Daily and monthly	Polystyrene latex suspension	5 times over 3 years
Wind speed by high sensitivity anemometer	±0.3 m/s	Certified wind tunnel	Factor calibration	When out of spec	Visual check, synchronized motor when out of spec	Daily and when needed	Synchronized motor	Yearly
Wind direction by wind vane	±10° from north	Surveyor compass, solar azimuth	Surveyor compass, solar azimuth	When out of spec	Visual check, realignment when out of spec	Weekly	Surveyor compass, solar azimuth	Yearly
Temperature by platinum resistance	±0.1 °C	NIST thermometer and water bath	Factory calibration	When out of spec	Psychrometer	Weekly	NIST thermometer and water bath	Yearly
Relative humidity by capacitance	±2%	NIST thermometer and dew cups	Factory calibration	When out of spec	Psychrometer	Weekly	Collocated chilled mirror	Yearly
Solar radiation by pyranometer	±20 w/m^2	NIST standard luminance	Factory calibration	When out of spec	Visual inspection of max and min	Weekly	Collocated pyranometer	Yearly
Mass by electrobalance	±10%	Class 1.1 weights	Class 1.1 weights	Each weighing session	Class 1.1 weights, replicates, and blanks	Every 10 samples	NIST Class 1.1 weights	Yearly
Total elements (X-ray fluorescence)	±5%	EPA polymer films, NIST impregnated glass	Micromatter film deposits	6 months or when out of spec	NIST impregnated glass, replicates, and blanks	Every 15 samples	Micromatter film deposits and interlaboratory comparison	Yearly
Soluble metals, anions, and cations by atomic absorption spectrometry, ion chromatography, and automated colorimetry	±0.005 to ±0.05 µg/mL	Mineral salt solutions	Mineral salt solutions	Each analysis session	Mixed salt solutions, replicates, and blanks	Every 10 samples	Mixed salt solutions and interlaboratory comparison	Yearly
Carbon by thermal evolution with transmission and reflectance pyrolysis correction	±0.2 µg/cm^2	NIST CO_2 and CH_4	Phthalate and sucrose solutions	3 months or when out of spec	CH_4 Replicates and blanks	Every sample Every 10 samples	Phthalate and sucrose solutions on filters CO_2 and CH_4 and interlaboratory comparison	Yearly
C_2–C_{12} hydrocarbons from canisters by cryofocusing, gas chromatography, and flame ionization	±0.2 ppbC	NIST CH_4	Certified CH_4 and reference gases	Each analysis session	Reference gases, replicates, and blanks	Every 10 samples	Interlaboratory comparison	Yearly
C_8–C_{20} hydrocarbons from TENAX by thermal desorption, gas chromatography, and flame ionization	±0.2 ppbC	NIST CH_4	Certified CH_4 and reference gases	Each analysis session	Reference gases, replicates, and blanks	Every 10 samples	Interlaboratory comparison	Yearly
C_8–C_7 carbonyls from DNPH by high pressure liquid chromatography and ultraviolet detection	±0.2 ppbC	Aldehyde mixtures	Aldehyde mixtures	Each analysis session	Reference mixtures, replicates, and blanks	Every 10 samples	Interlaboratory comparison	Yearly
Particulate organic compounds from carbonyls from DNPH by high pressure liquid chromatography and ultraviolet detection	±0.2 µg/m^3	Compound mixtures	Compound mixtures	Each analysis session	Reference mixtures, replicates, and blanks	Every 10 samples	Interlaboratory comparison	Yearly

are used to determine when an instrument is mis-performing as well as to estimate the precision of each measurement. Audits are used to evaluate the traceability and the accuracy of the measurements.

Each in-situ instrument is interfaced to an analogue and serial data acquisition system that is accessible by modem for data review and archiving. This communication capability will be used by health researchers to time clinical and toxicological measurements of test subjects.

The Supersite data base is available over the Internet where it is integrated with other particulate, gaseous, visibility, and meteorological data bases throughout central California and with measurements from other Supersites.[43] The following types of tables are included in the project database:

- Measurement locations: Each measurement location is identified with a unique alphanumeric site ID accompanied by its name and address, coordinates, elevation, primary operator, and a summary of measurements taken at the site for different monitoring periods. Coordinates are determined with global positioning systems (GPS) using map basis NAD-83 (Federal Aviation Administration convention). The GPS time stamp is recorded to correct coordinate deviations.
- Variable definitions: Each variable is assigned a unique code that is accompanied by its definition, units, averaging time, measurement method, applicable temperature and pressure adjustments, and data reporting format.
- Data validation flags: Flags specific to each measurement are translated into a common set of validation flags that are carried with each data point. These are currently being defined by EPA for its speciation program, and this will be a starting point for Fresno Supersite data validation flags.
- Data tables: Basic data tables are constructed in normalized formats that have the same structure for different types of data. Each record contains the site code, sample date (MM/DD/YYYY), sample time (HH:MM:SS PST), variable code, measurement value, measurement precision, validity code, and validation level. These files will be transparent to most users and can be easily manipulated into convenient data analysis forms. Missing or invalid measurements are omitted or contain a "NULL" value. Separate tables are produced for different averaging times and for non-uniform data sets.
- Validation indicators: Detailed information on specific samples indicating the nature of the data qualification. These tables also contain the validation level assigned to each data item.

Data validity levels are designated in the validation tables for different stages of data acquisition and interpretation. Level 0 designates data sets downloaded from a field instrument that have not been examined. These measurements are used to evaluate instrument performance and to forecast conditions for special experiments, but they are not used for interpretive purposes.

Level 1 data have been evaluated by the measurement investigator prior to submission to the database. Values are removed for instrument downtime and performance tests, adjustments for calibration deviations are applied, extreme values are investigated, internal comparisons are made, blanks are subtracted, precisions are estimated and propagated, and appropriate data qualification flags are assigned. For sequential measurements, jump tests, standard deviation tests, and extreme value tests often identify values that need to be investigated.

Level 2 data have completed comparison tests between data sets. These tests often result in the investigation of several samples that do not follow the same pattern as other measurements. Samples are sometimes reanalyzed and re-designated as valid, invalid, or suspect as a result of the investigation.

Level 3 validation occurs after measurements are used to test hypotheses and after values that are found contradictory to other values have been investigated. The first assumption upon finding a measurement inconsistent with physical expectations is that the unusual value is due to a measurement error. If, upon tracing the path of the measurement, nothing unusual is found, the value can be assumed to be a valid result of an environmental cause. Unusual values are identified as 1) extreme values; 2) values that would normally track the values of other variables in a time series; and 3) values for observables that would normally follow a qualitatively predictable spatial or temporal pattern.

HYPOTHESES AND TESTING METHODS

Table 3 summarizes the hypotheses associated with each objective. These hypotheses take advantage of the longer-than-3-year record that will be available. $PM_{2.5}$ and PM_{10} NAAQS require at least a 3-year record to determine attainment.[44] While the CRPAQS monitoring and data analysis will provide detailed understanding of source contributions, meteorology, and atmospheric chemistry for 1 year and several multi-day episodes, they cannot provide the multi-year perspective needed to evaluate compliance with current NAAQS. The application and evaluation of advanced, continuous measurement technology over this extended period will provide knowledge needed by regulators when they consider particle health indicators and candidates for future FRMs.

Table 3. Hypotheses to be tested by Fresno Supersite measurements.

1. Measurement Method Evaluation Hypotheses

1.1 Mass and chemical (elements, ions, and carbon) measurements from compliance filter samplers with Teflon-membrane filters (from FRM or Minivol monitors) represent $PM_{2.5}$ mass within the spatial zone of representation of the Fresno Supersite. Community representative sites that meet $PM_{2.5}$ siting criteria are intended to approximate the exposure of many people to $PM_{2.5}$ in outdoor air, but there are few empirical measurements to verify that these criteria are adequate for this objective.

1.2 Elemental analysis of Teflon-membrane filters under helium and vacuum atmospheres does not result in a significant (>10%) loss of volatile NO_3^-. Volatile species, such as NH_4NO_3, may experience less evaporation when analyzed in helium than under vacuum. If so, then FRM Teflon filters might also be used after XRF for NO_3 analysis.

1.3 Gaseous organic carbon absorbed on quartz-fiber filters is a small (<15%) fraction of organic carbon measured on these filters. Carbon measurements taken at other sites without denuders and backup filters can be considered more precise if this is proven.

1.4 Volatilized particulate NO_3^- is a minor (<10%) part of particle NO_3^- during winter, but a major fraction of particle NO_3^- during other seasons. NO_3^- measurements taken at other sites without denuders and backup filters can be considered more precise if this is proven.

1.5 Volatilized particulate NO_3^- is a minor (<10%) part of actual $PM_{2.5}$ during all seasons. Although larger fractions of NO_3^- may be lost during sampling for non-winter periods, this may be a small fraction of $PM_{2.5}$ mass.

1.6 $PM_{2.5}$ mass concentrations estimated from particle size, weighted sums of chemical components, light scattering, light absorption, and light extinction are equivalent to gravimetric mass of samples taken with a $PM_{2.5}$ FRM sampler. Knowing the conditions under which equivalence can be expected will expand the utility of different continuous $PM_{2.5}$ surrogate measurements taken at other sites and lessen the need for extensive collocated measurements.

1.7 $PM_{2.5}$ and PM_{10} mass measurements are comparable for all measurement methods during spring and summer when the sampled aerosol is stable. Mass measurements diverge during winter and part of the spring when volatile NO_3^- and organics constitute a large mass fraction. This is especially relevant to measurement methods that heat the sample, such as the TEOM.

1.8 Particle number counts in integrated sub-ranges of the 0.01 to ~10 μm size distribution are comparable to $PM_{2.5}$ and PM_{10} mass measurements assuming constant particle shape and density. Variable particle compositions may cause this assumption to be incorrect.

1.9 Differences between continuous PM_{10} and $PM_{2.5}$ monitors are comparable to coarse particle mass concentrations on a dichotomous sampler. If biases in BAM and TEOM measurements affect the PM_{10} and $PM_{2.5}$ samples in the same way, the coarse mass fraction may be comparable to that of a dichotomous virtual impactor filter measurement.

1.10 The $PM_{2.5}$ geological component is comparable to the difference between continuous $PM_{2.5}$ mass measurements and the sum of continuous NO_3^-, SO_4^{-2}, and carbon concentrations (adjusted for ammonium, hydrogen, and oxygen). If this holds true for the short-term measurements, then the geological component could be inferred from the other continuous measurements, providing complete characterization of major components at ~30 min intervals.

1.11 Bioaerosols and endotoxins constitute a constant fraction of coarse particle mass. Aluminum, silicon, calcium, titanium, iron, and other elements have reasonably uniform proportions in soils. Bioaerosols (e.g., pollen, molds, spores) and endotoxins are less likely to exhibit constant proportions to coarse particle mass and elemental concentrations.

1.12 Photoionization and wavelength-specific light absorption are correlated with organic compound concentrations. These methods have shown good correlations with aggregates of certain organic compounds. Simultaneous PAH and other organic samples substances can be used to evaluate this relationship.

2. Aerosol Characterization and Control Strategy Hypotheses

2.1 Short duration (~5 min) spikes in particle measurements represent contributions from nearby (<500 m) emitters. If true, zones of influence for specific emitters can be determined from measurements at a single site rather than multiple sites.

2.2 Nearby emitters represent a small (<15%) fraction of $PM_{2.5}$ measured at a community representative sampling site. Spikes caused by nearby emitters should be a small proportion of the $PM_{2.5}$ contributed by sources with urban- and regional-scale zones of influence at a community representative monitor.

2.3 The majority of ultrafine particles (<0.1 μm) are from nearby (<500 m) fresh emission sources. Small particles with aerodynamic diameters <0.1 μm coagulate with larger particles within a short distance of their emission. If this is true, their numbers should increase in short duration spikes and be consistent with wind directions favoring nearby emitters, such as highly traveled First Street.

2.4 NH_4NO_3 reductions are limited by available HNO_3 rather than available NH_3 in urban areas during all seasons and all hours of the day. Diurnal measurements of NH_3, HNO_3, SO_4^{-2}, and NO_3^- have not been available to evaluate this limitation.

2.5 Advanced gas and particle organic speciation measurements, coupled with elements, ions, and organic and elemental carbon fractions, consistently and accurately distinguish contributions from different types of suspended dust, secondary SO_4^{-2} and NO_3^-, vegetative burning (wood and field burning and meat cooking), gasoline engine exhaust (cold starts, high emitters, and hot stabilized operations), diesel exhaust, and primary industrial emissions. Source contributions will be estimated using receptor models with these data as inputs. Source profiles are being measured in separate studies.

2.6 Commonly measured elements, ions, and organic and elemental carbon fractions consistently and accurately distinguish contributions from suspended dust, secondary SO_4^{-2} and NO_3^-, vegetative burning (wood and field burning and meat cooking), gasoline engine exhaust (cold starts, high emitters, and hot stabilized), diesel exhaust, and primary industrial emissions. Specificity of source contributions is limited without detailed organic speciation, but general source categories might still be apportioned. Receptor model results will be compared with those of the detailed organic apportionments.

2.7 Gasoline engine cold starts and high emitters are the major causes of gasoline-fueled vehicle contributions to $PM_{2.5}$, and they cause gasoline exhaust contributions to exceed diesel exhaust contributions. If this is true, then current inspection and maintenance programs may need to be modified.

2.8 Statistical aggregates of concentration indicators for a single year deviate by less than sampling error from a 3-year distribution. If this is true, then shorter-term studies can be used to reliably represent the 3 years needed to determine compliance with $PM_{2.5}$ NAAQS.

2.9 Concentrations in continuously measured carbon fractions can be associated with different proportions of gasoline vehicle exhaust, diesel vehicle exhaust, and wood burning. Ratios of PAH, carbon, and light absorption measurements may differ for different sources and allow their discrimination at receptors.

2.10 Annual average, seasonal average, and 98th percentile source contributions from fugitive dust, wood burning and cooking, vehicle exhaust, secondary $(NH_4)_2SO_4$, and secondary NH_4NO_3 differ by less than ±10% over a 3-year

Table 3. Continued.

	period. Source contributions determined in a short-term study can be extrapolated to a multi-year period.
2.11	Indicators of particle concentrations (e.g., particle size fraction, number, surface area, and major chemical component) are highly correlated; one indicator is equivalent to other indicators that might be specified by future air quality standards.
2.12	Large reductions in $PM_{2.5}$ and PM_{10} mass concentrations in central California after 1992 are due to the end of a 7-year drought rather than due to emissions reductions. Many of the highest $PM_{2.5}$ and PM_{10} were measured during the drought. Comparisons will be made between data from these early periods and the Supersite monitoring period.
3.	**Health- and Exposure-Related Hypotheses**
3.1	$PM_{2.5}$ mass concentration, surface area, and number counts are highly correlated ($R^2 > 0.8$), and a measure of one is a good indicator of the other two. High correlations demonstrate collinearity that makes epidemiologic relationships to individual variables uninterpretable.
3.2	The soluble fraction of transition metals, especially those with specific valence states such as Cr(VI) and Fe(III), which are believed to have adverse health effects, is a small fraction (<15%) of total metal concentrations in $PM_{2.5}$. This fraction may not be accurately quantified by total elemental analysis methods that are usually applied to aerosol samples.
3.3	Measurements at a community representative sampling site represent the minimum to which people are exposed in their neighborhoods within an urban area. Nearby emitters, such as wood stoves and fireplaces or freeways, may add to outdoor exposure levels. Satellite site measurements are taken in these areas for comparison with Supersite data.
3.4	Hospital and physician diagnoses of respiratory and cardiovascular ailments are equally as sensitive to changes in $PM_{2.5}$ mass concentrations as they are to other air pollution indicators. A variety of indicators will be available at Fresno that can be associated with different end-points.
3.5	There is a discernible lower threshold for single and combined air quality indicators, below which no relationships with health end-points are statistically significant. More precise and spatially homogenous measurements will allow a more precise examination of this threshold for different aerosol characteristics and exposure periods.
3.6	Different particle chemical characteristics have different and identifiable immuno-enhancing properties that affect the symptom onset and severity of short-term reductions in lung function, asthma attacks, and cardiovascular ailments. Asthma in children may be aggravated by elevated levels of endotoxins, bioaerosols, PAH, or diesel exhaust to a greater extent than it is by particle number, surface, or volume concentrations.
3.7	Coarse particle concentrations and biologically active components show relationships to health end-points. Coarse particles are not as highly correlated with $PM_{2.5}$, and they may have different and detectable harmful effects.
3.8	Animal (rats) exposures to different combinations of concentrated amounts of particle size, surface area, chemical, and mass characteristics result in similar indications of respiratory and cardiovascular distress. Aerosol concentrators are used to vary the exposure of test subjects to a variety of aerosol compositions and levels. Reactions are often different from laboratory-generated aerosols.
3.9	Particles found in healthy human lungs have characteristics similar to those found in urban air over long periods of exposure. Examination of autopsied lungs from healthy Fresno residents after accidental death shows particulate accumulations.

Measurement methods evaluation hypotheses (1.1–1.12) are based on evidence that there is a climatology for the validity and comparability of measurements acquired by the same instruments. Meteorological conditions, source contributions, and aerosol chemical composition in central California are known to change substantially over a year and even between different parts of the day.[5-10] Long-term measurements for a year or more are needed to evaluate the feasibility, practicality, and equivalence of different measurement techniques. Evaluation of these measurements will determine where and when less complex, more convenient, or more widely available measurements can be used in place of the advanced methods implemented at the Supersite.

Several empirical and statistical measures are applied to evaluate predictability and equivalence.[45] Linear regression is most commonly used and is the requirement for Federal Equivalent Method relationships with FRMs. Other comparison measures include average ratios and standard deviations, ratios of averages, and the distribution of differences (X minus Y) for <1s, 1s to 2s, 2s to 3s, and >3s precision intervals. These measures indicate the extent to which long-term averages are more or less equivalent than individual values and whether or not the majority of differences are within stated uncertainty intervals.

Emissions reduction plans need to determine source contributions to primary particles and the limiting precursors for secondary particles. Conclusions drawn from special, short-term studies need to be generalized over at least the NAAQS 3-year period, and over a longer period (~10 years) during which control strategies are implemented. Hypotheses 2.1–2.12 examine how well conclusions from special studies such as CRPAQS stand up during subsequent years. They also place the Supersite monitoring period within a long-term record by comparison with historical $PM_{2.5}$, PM_{10}, light scattering, coefficient of haze, and meteorological measurements from 1990 onward.

Figure 1 shows an example of how different measurements varied over the course of Christmas day, 1999. There are clear variations in absolute and relative concentrations that are probably related to different emissions, vertical mixing, and horizontal transport that change throughout the day. The mass indicators exhibit maxima during both midday and late evening. The compositional data show that the midday maximum coincides with the NO_3^- maximum while the late-evening peak coincides with increased black carbon (BC) and PAHs. The PAH to BC ratio changes throughout the day, with a substantial decrease in the afternoon. Since PAH is highly reactive, especially in sunlight, this may indicate an aged rather than fresh source of BC. These variations have implications for understanding sources and interpreting health data.

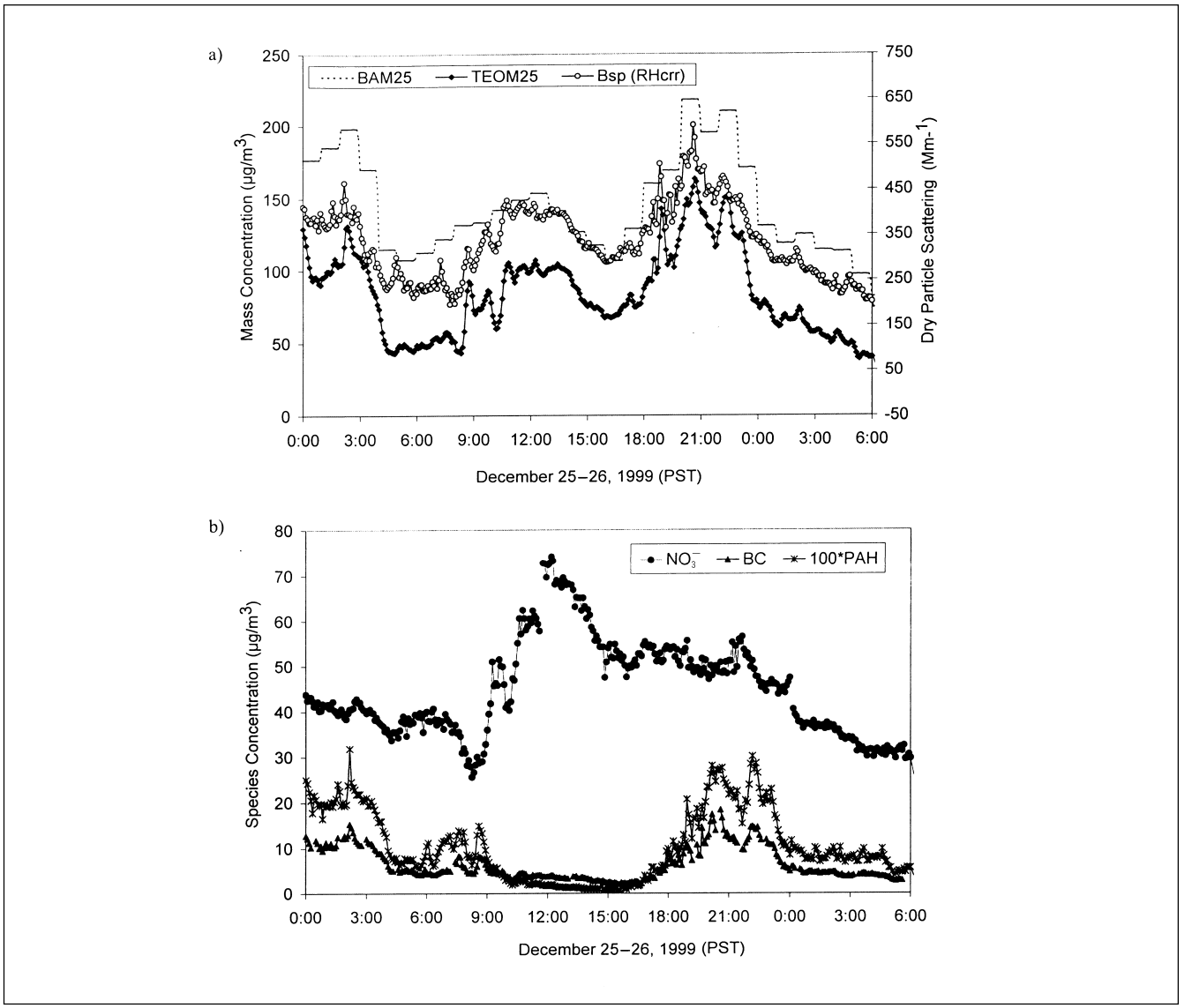


Figure 1. Example of 5-min and hourly measurements of $PM_{2.5}$ for December 25, 1999, for a) $PM_{2.5}$ mass from beta gauge (BAM25), tapered element oscillating microbalance (TEOM25), and particle light scattering from a nephelometer corrected for particulate water based on the measured relative humidity (Bsp-RHcrr); and b) 5-min concentrations profiles for $PM_{2.5}$ NO_3^-, black carbon (BC), and PAH. Note the different diurnal resolutions of BC and NO_3^-. BC and PAH are higher during the morning and at night, while NO_3^- peaks during the afternoon. The $PM_{2.5}$ mass shows afternoon and nighttime increases. The $PM_{2.5}$ TEOM was operated at 30 °C with a heatless drier in this example, but there is still evidence of NO_3^- volatilization.

Mauderly et al.[46] identify several indicators for adverse health effects: 1) PM mass; 2) PM surface area; 3) PM number (i.e., ultrafine concentration); 4) transition metals (especially the soluble fraction); 5) acids (especially sulfuric acid); 6) organic compounds; 7) bioaerosols; 8) SO_4^{-2} and NO_3^- compounds (typically neutralized in whole or in part by NH_3 or sodium); 9) peroxides and other free radicals that accompany and help to form PM; 10) soot (elemental carbon and associated PAH); and 11) correlated co-factors (other pollutants and variation in meteorology). Long-term data records of these variables are needed to examine relationships to health end-points and to determine the range of concentrations to which humans might be exposed. Owing to the complexity and expense of measurement technology, such long-term records are lacking.

Measurements at the Fresno Supersite can support health studies related to all but category 9, peroxides and free radicals. Although H_2SO_4 (category 5) could be quantified, there is sufficient evidence that available sulfate anions are completely neutralized by NH_3 and alkaline species in central California. HNO_3 could be related to health endpoints, and attempts at its continuous measurement are needed.

Hypotheses 3.1–3.9 in Table 3 require measurements from other studies related to human respiratory health in the Fresno area and central California.[47] The Particulate

Air Pollution and the Natural History of Childhood Asthma Study sponsored by ARB at UC Berkeley will test panels of children for 4 years, with a variety of lung function tests and clinical examinations made throughout the period. Indoor and outdoor samples will be acquired and personal exposure monitors will be used to develop long-term exposure estimates. These will be correlated with Supersite measurements.

The Health Effects of Concentrated Ambient Particles from the Central Valley of California sponsored by EPA at UC Davis will expose rats to different levels of Central Valley aerosol, then sacrifice the rats and examine damage to their respiratory system. Portable particle concentrators for ultrafine, fine, and coarse fractions will be located near the Supersite to take advantage of its measurements. Extremes will be sought in particle number, composition, surface area, and other variables by selecting times of day and times of the year where contrasts are largest. Real-time access to Supersite measurements will be used to schedule experiments.

The Indoor Exposure from Ambient Concentrations Study sponsored by the U.S. Department of Energy at Lawrence Berkeley National Laboratory (Berkeley, CA) will develop a mechanistic model for infiltration of outside air into buildings. The model requires a detailed understanding of particle size and chemical characteristics, infiltration characteristics of various buildings, resident behavior, and outdoor meteorological conditions. Indoor and outdoor measurements will be taken for representative seasons to test the model. Supersite measurements will be used for detailed characterization and to extrapolate limited and integrated indoor/outdoor measurements to a range of aerosol characteristics and weather conditions.

SUMMARY

The Fresno Supersite will provide 1) a long-term record of simultaneous advanced particle measurements that includes a large range of concentration levels, particle sizes, and aerosol compositions, suitable for many purposes; 2) supportable conclusions about specific hypotheses concerning measurement method performance, causes of excessive pollution levels, and effects on health; 3) linkages and collaboration among air quality scientists, toxicologists, epidemiologists, exposure specialists, and clinicians that better integrate and communicate their scientific findings; 4) a research infrastructure in Fresno and at the First Street site to support other studies during and after Supersite monitoring; and 5) publications and reports that support local, state, and national decision-making related to standard setting and pollution controls.

Subsets of Fresno Supersite measurements will be comparable to similar measurements from other Supersites established nationwide and to other measurements from additional sites and special studies in central California. Specifically, the non-urban Angiola Tower site will have nearly identical measurements that will allow differentiation of regional from urban contributions.

Fresno Supersite measurements will establish relationships between complex and less comprehensive measurements such as those acquired from speciation monitors, FRMs, and Minivol monitors. These data will be available over a wider spatial scale, both in central California and throughout the United States. A better understanding of where and when these more widely available measurements are adequate surrogates for more complex measurements will provide opportunities for comparisons and contrasts.

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